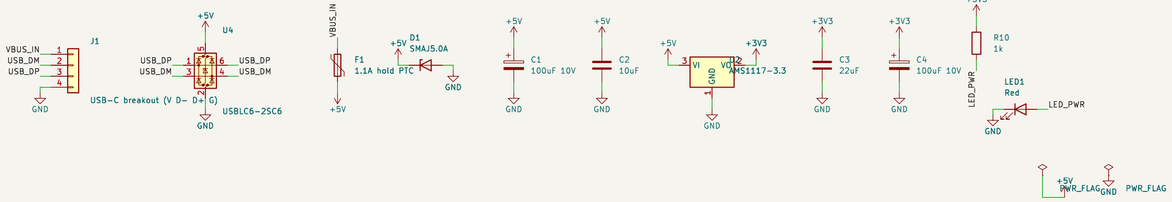
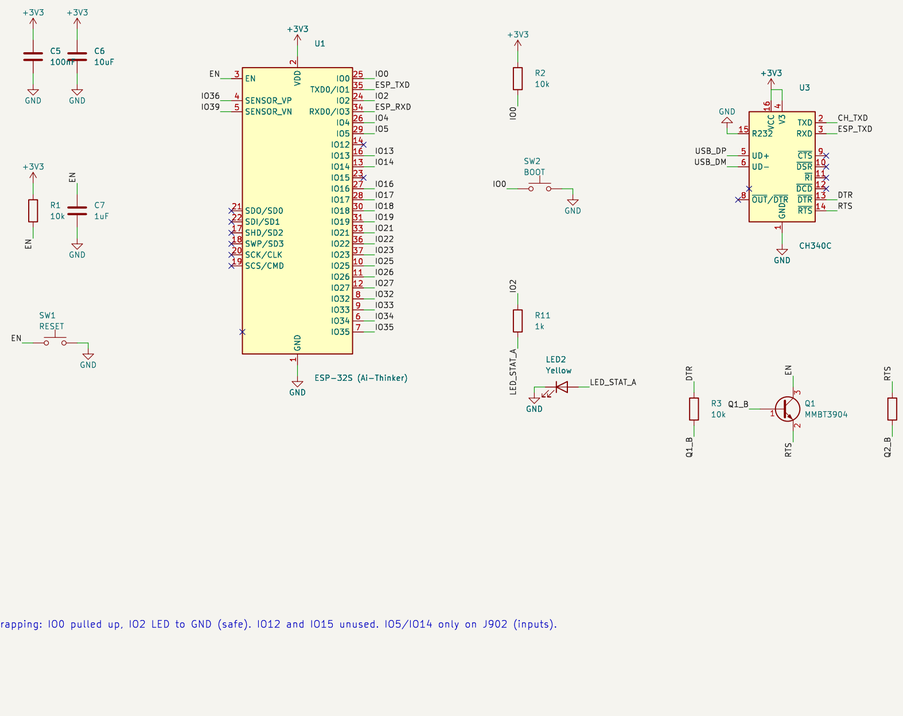
PORTS: XH 4P = 1 GND, 2 VP, 3 A, 4 B. VP = 3V3 or 5V by solder jumper (pin 1 side bridged). 0.5 A PTC per port.

POWER: USB-C breakout -> F1 1.1 A PTC -> D1 TVS -> +5V -> AMS1117-3.3. USB data via USBLC6 ESD to CH340C.

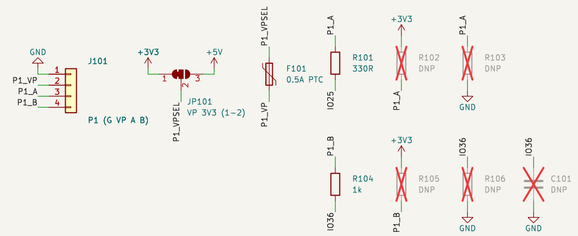


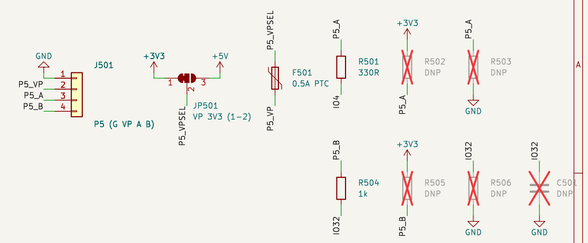
MCU + USB-UART: CH340C in 3.3 V mode (V3 tied to VCC). DTR/RTS auto-reset. IO0 = BOOT, IO2 = status LED.

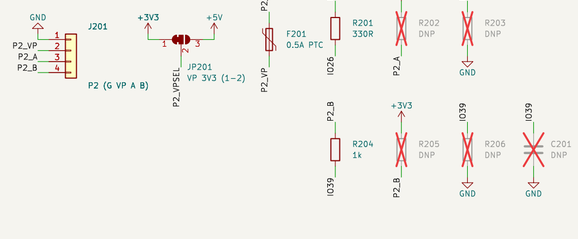


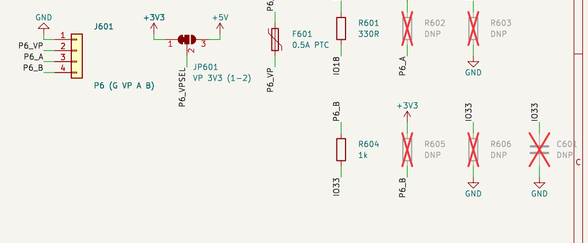
Strapping: IO0 pulled up, IO2 LED to GND (safe). IO12 and IO15 unused. IO5/IO14 only on J902 (inputs).

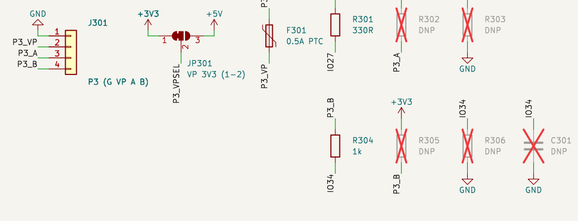


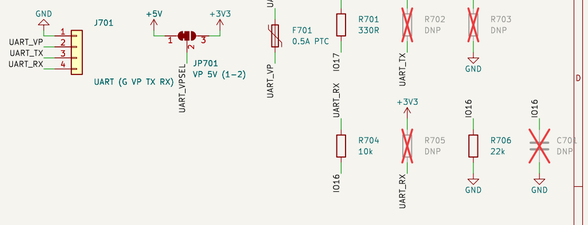


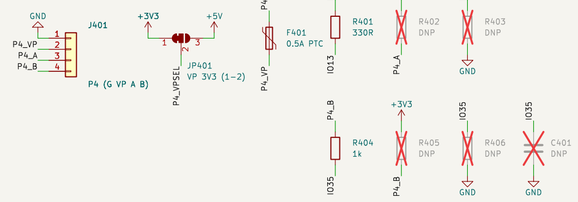


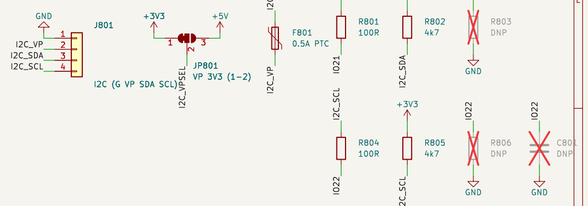






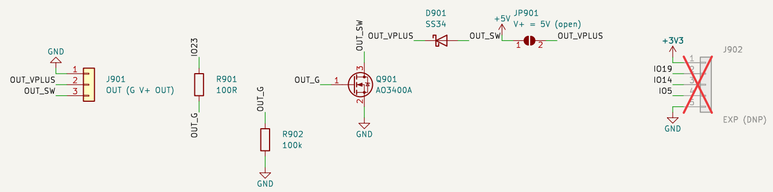






Rn01 series A (330R) | Rn02 A pull-up | Rn03 A pull-down | Rn04 series B (1k) | Rn05 B pull-up | Rn06 / Cn01 B to GND (GPIO side). DNP = option footprint.

OUT: AO3400A low-side switch on IO23, SS34 flyback to V+. JP901 ties V+ to +5V (open by default).



6 generic ports + UART + I2C + switched output. USB-C 5 V, CH340C flashing.

Sheet: /
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Title: ESP32 Modular Base

Size: A2	Date: 2026-10-13	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1